

Alexandre LOURDEAUX

PhD in Mathematics

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Current situation

09/2024–08/2026 **Postdoctoral fellow**, *Southern University of Science and Technology*, Shenzhen, China.
Mentors : Yong Hu, Efim Zelmanov

Degrees and studies

09/2023–08/2024 **ATER (~Postdoctoral position)**, *Université de Grenoble-Alpes*, Grenoble, France.

09/2024–08/2026 **Postdoctoral fellow**, *University of Alberta*, Edmonton, Canada.
Mentors : Arturo Pianzola, Vladimir Chernousov

01/2021–06/2021 **ATER (~Postdoctoral position)**, *University of Franche Comté*, Besançon, France.

2020 **PhD in mathematics**, *University Claude Bernard Lyon 1, Camille Jordan Institute*, Lyon, France.

Title : On Cohomological invariants of linear algebraic groups.

Advisor : Philippe Gille

Referees : Stefan Gille, Jean-Pierre Tignol.

Jury : Anna Cadoret, Frédéric Déglise, Kenji Iohara, Anne Quéguiner-Mathieu, Jean-Pierre Tignol.

Janv. 2017 **Diploma of École Normale Supérieure de Paris.**

2015 **Master degree in fundamental mathematics**, *University Paris 6*, Paris, France.

2011–2012 **Bachelor degree in mathematics**, *University Paris 7*, Paris, France.

2011–2016 **Student at École Normale Supérieure de Paris**, Paris, France.

Experiences

10/2021–10/2023 **Postdoctoral Fellow**, *University of Alberta*, Edmonton, Canada.

Advisors : Arturo Pianzola & Vladimir Chernousov.

Fév.-Mars 2020 **Invited searcher-student**, *University of Ottawa*, Ottawa, Canada.

Supervisor : Kirill Zaynullin.

Teaching

2023–2024 **Various exercise sessions for undergraduates, and Instructor of 'dynamical systems' for biologists**, *Grenoble University*.

2023 **Instructor for 'Linear Algebra II'**, *University of Alberta*.

- 2022 **Instructor for 'Intermediate Calculus II'**, *University of Alberta*.
- Jan.–June 2021 **Various exercise sessions for math undergraduates, and Instructor of 'differential equations' for biologists**, *University of Franche Comté*.
- 2017–2019 **Various exercise sessions for math undergraduates**, *Lyon 1 University*.

Skills

Computer

Languages Latex, Caml, C++

Languages

French Native language

English Professionnal level

Mandarin A2/HSK2 level

Publications

On degree 2 cohomological invariants of liner algebraic groups, *Journal of Pure and Applied Algebra*, vol. 226 (10), Article 107059, 2022.

On geometry of pseudo-reductive groups, *Communications in Algebra*, vol. 50 (12), pp. 5371-5386, 2022.

Preprints

(With Anis Zidani, submitted) *Conjecture I for unirational algebraic groups over imperfect fields*, *arXiv :2604.05148*, 2026.

(With Yong Hu) *Cohomological invariants of hermitian forms that detect hyperbolicity*, *arXiv :2604.00581*, 2026.

Faisceaux $Q/Z(j)$ et conjecture de Gersten sur un corps imparfait, *arXiv :2010.09182*, 2020.

In preparation

(With Yong Hu) *Hasse principle for hermitian lattices*.

Talks

- 2026 **Seminar 'Number Theory'**, *IMRA*, Bucharest (Romania).

Title : On a Serre's Conjecture I for unirational groups

Great Bay University seminar, *Great Bay University*, Dongguan (China).

Title : Conjecture I for unirational groups

- 2025 **Seminar 'Online Math Forum'**, *Westlake University*, Hangzhou (China).

Title : Brauer invariants of linear algebraic groups

- 2024 **Research minicourse**, *Southern University of Science and Technology*, Shenzhen (China).

Subject : Pseudo-reductive groups

Seminar 'Séminaire Compréhensible', *Grenoble University*, Grenoble (France).

Title : Quadratic forms over a field

- Seminar 'GAAO'**, *Clermont-Auvergne University*, Clermont-Ferrand (France).
 Title : Brauer invariants of linear algebraic groups
- 2023 **Seminar 'Algèbre et géométrie'**, *Université de Grenoble*, Grenoble (France).
 Title : Invariants de Brauer des groupes algébriques linéaires
- 2022 **Conference 'Groups, Rings, Lie, and Hopf algebras IV'**, *Memorial University of Newfoundland*, St John's (Canada).
 Title : Brauer invariants of linear algebraic groups.
- 2020 **Online seminar "Quadratic forms, Linear algebraic groups and Beyond"**.
 Brauer invariants of linear algebraic groups.
- 2020 **Algebra seminar**, *University of Ottawa*, Ottawa (Canada).
 Title : On cohomological invariants of linear algebraic groups.
- 2019 **Algebra Seminar**, *Lyon 1 University*, Lyon (France).
 Title : Sur les invariants cohomologiques des groupes algébriques linéaires.